

Chapter 7:

Work and Kinetic Energy

Work and Energy



Chapter-7: Outline

7.1 Forms of Energy

7.2 Work Done by a Constant Force

7.3 Work Done by a Varying Force

7.4 Kinetic Energy

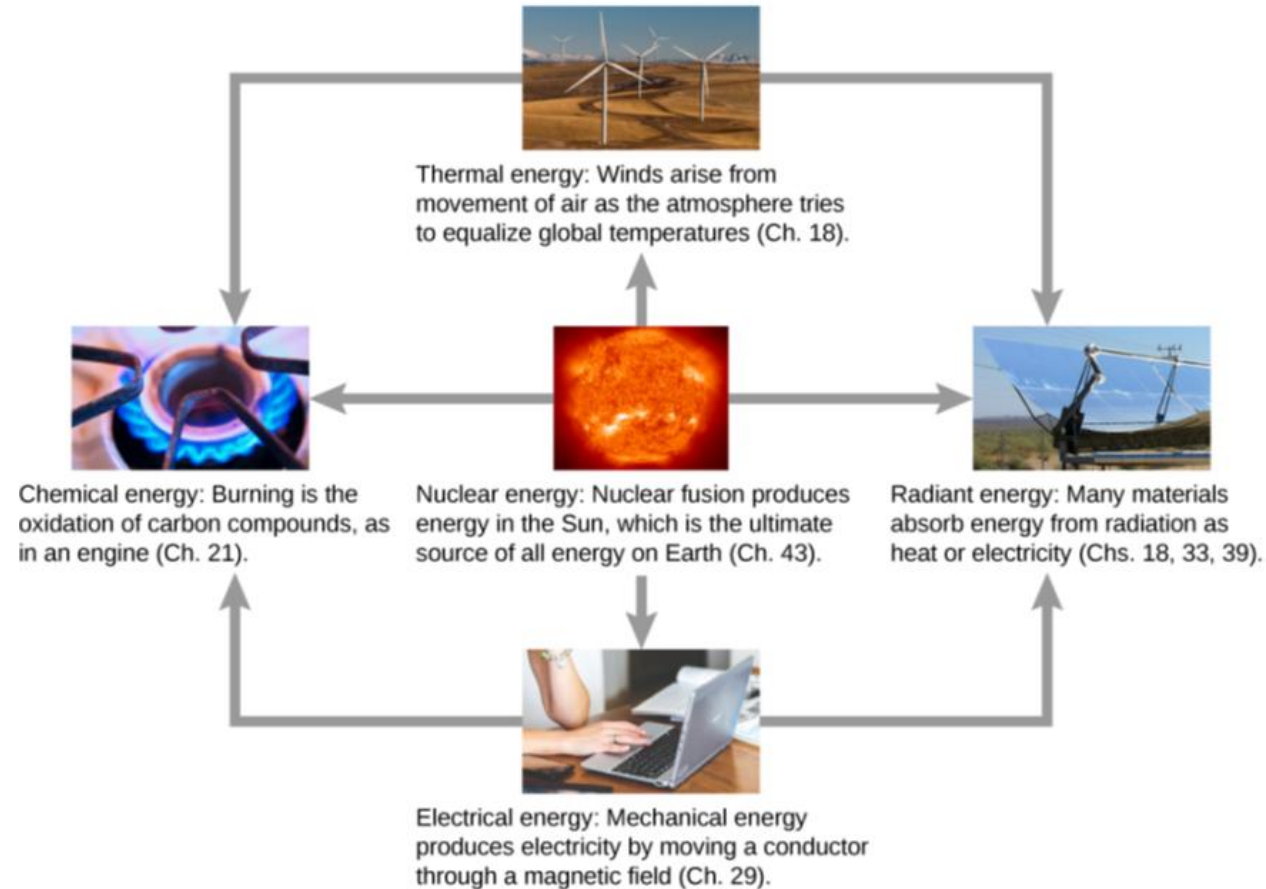
7.5 Work–Energy Theorem

7.6 Power

Helpful Videos

- Work and Energy:
<https://www.youtube.com/watch?v=zVRH9d5PW8g>
- Kinetic Energy and Potential Energy:
<https://www.youtube.com/watch?v=g7u6pIfUVy4>
- How do energy convert?
<https://www.youtube.com/watch?v=ftj23FRS2LI>
- Physics: Energy Transformation:
<https://www.youtube.com/watch?v=O4Rxb-3jIqQ>
- Renewable Energy 101 – National Geographic:
<https://www.youtube.com/watch?v=1kUE0BZtTRc>

7.1 Forms of Energy



- Energy that we use in society takes many forms, which be converted from one into another depending on the process involved. We will study many of these forms of energy in later chapters of physics.

Forms of Energy

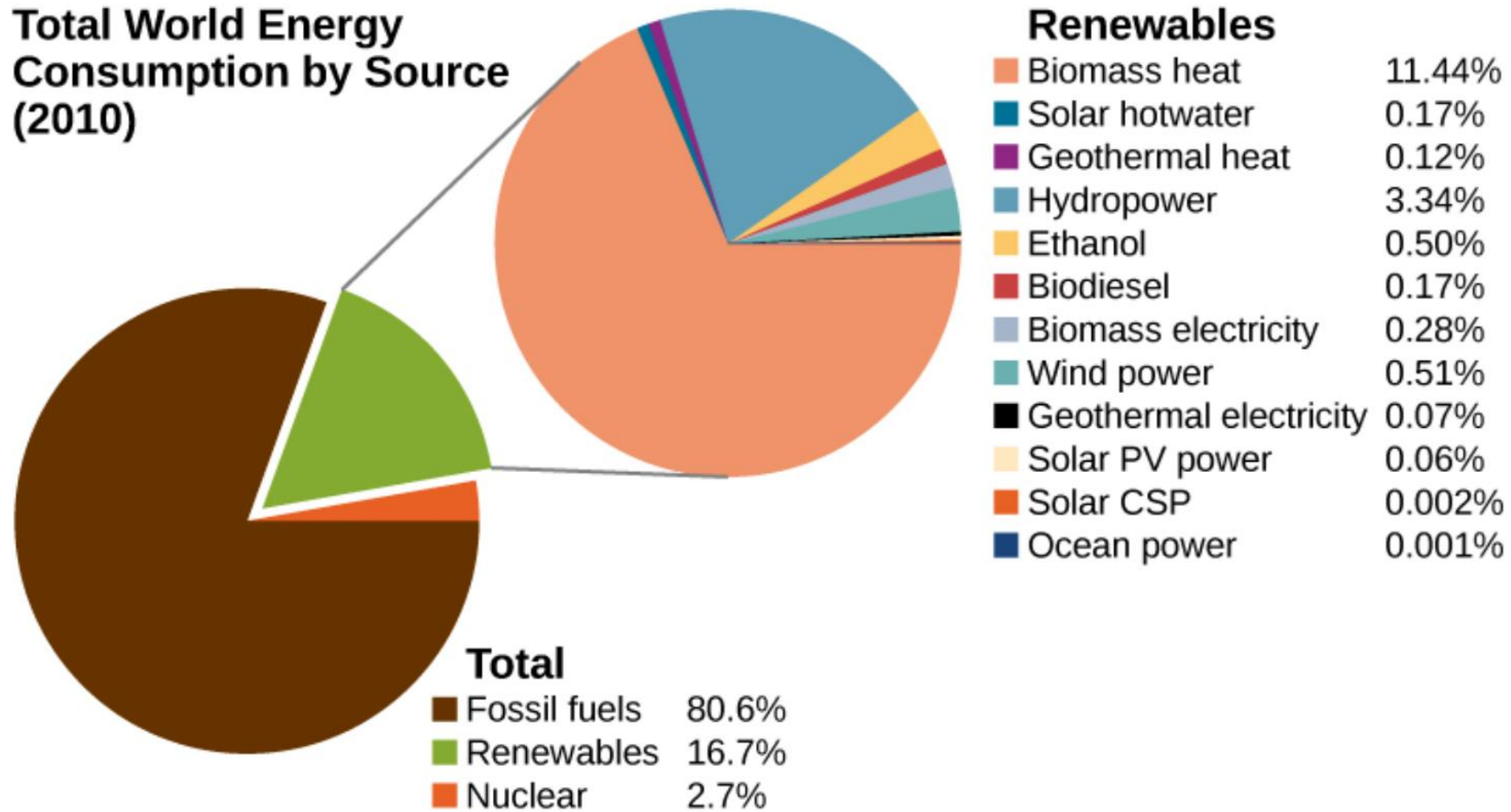
- Fuels, such as gasoline, have **chemical energy**, which is potential energy arising from their molecular structure. Chemical energy can be converted into thermal energy by reactions like oxidation.
- **Electrical energy** is a common form that is converted to many other forms and does work in a wide range of practical situations.
- Atoms and molecules inside all objects are in random motion. The internal kinetic energy from these random motions is called **thermal energy**, because it is related to the temperature of the object.
- Light is just one kind of electromagnetic radiation, or **radiant energy**, which also includes radio, infrared, ultraviolet, X-rays, and gamma rays. All bodies with thermal energy can radiate energy in electromagnetic waves.
- **Nuclear energy** comes from reactions and processes that convert measurable amounts of mass into energy. Nuclear energy is transformed into radiant energy in the Sun, into thermal energy in the boilers of nuclear power plants.
- These and all other forms of energy can be transformed into one another and, to a certain degree, can be converted into **mechanical work**.

How do energy convert?



Work and Energy

**Total World Energy
Consumption by Source
(2010)**



Renewable Energy



- Solar cell arrays found in a sunny area converting the solar energy into stored electrical energy.

Forms of energy will we study in our course?

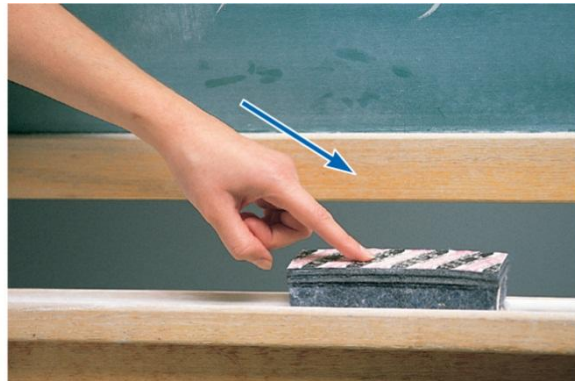
- We will mainly focus on energy that is associated with mechanical motions, or **mechanical energy**.
- Particularly, we will define **kinetic energy**, **potential energy**, and **work** as an energy transfer.
- We will study the **conservation of energy** and solve problems using the law of conservation of energy.
- The knowledge of energy provides another important way of solving mechanical motion problems.

7.2 Work Done by a Constant Force

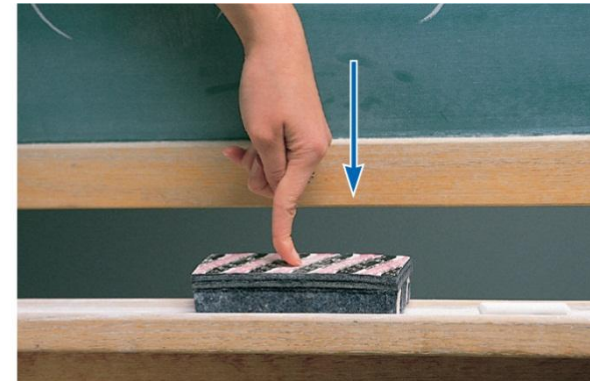
- **Work** : meaning in physics distinctly different from everyday meaning.
- In physics, **work** is done on an object when energy is transferred to the object. In other words, work is done when a **force** acts on something that undergoes a **displacement** from one position to another.
- Forces in three different directions: push applied in (b) more effective in moving eraser than push in (a) and not moving in (c).
- When analyzing forces to determine their influence on a system, must consider the **vector nature and magnitude of force**, and **magnitude of displacement**.



a



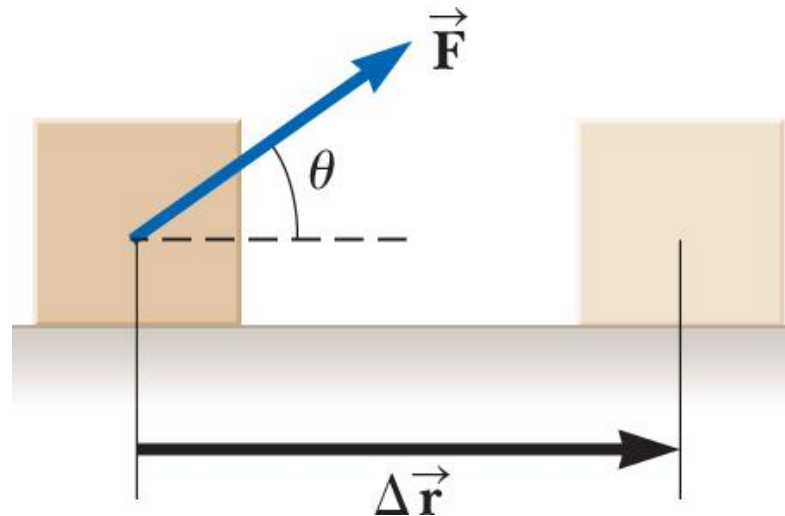
b



c

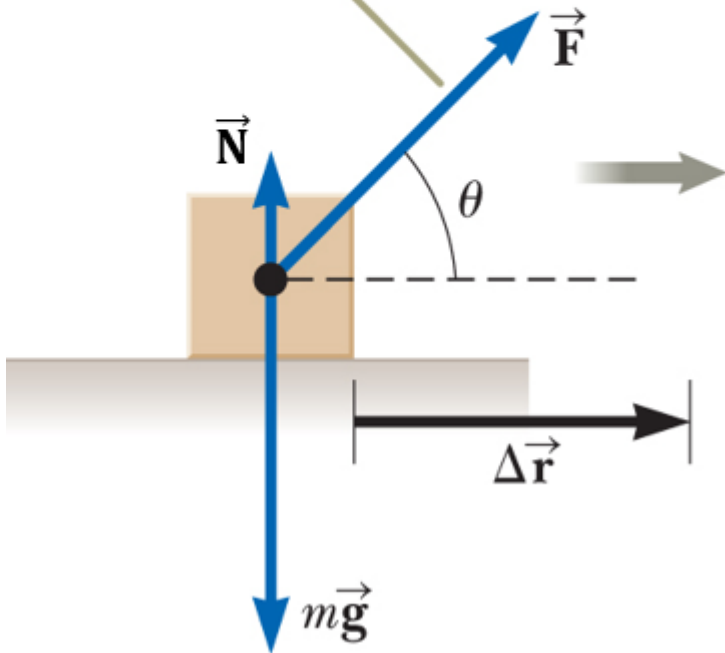
Work Done by a Constant Force

- **Definition:** Work W (scalar quantity) done on a system by an agent exerting a constant force on the system is the product of magnitude F of force, magnitude Δr of displacement of point of application of force, and $\cos \theta$:
- θ = angle between force and displacement vectors
- Force $\vec{F}$ and displacement $\Delta \vec{r}$ (*point of application of the force*)



Work Done by Constant Force

$\vec{F}$ is the only force that does work on the block in this situation.



$$W \equiv F \Delta r \cos \theta$$

- If $\theta = 0^\circ$, then $W = F \Delta r$.
- If $\theta = 90^\circ$, then $W = 0$, because $\cos 90^\circ = 0$.
- **Sign** of work depends on the directions of $\vec{F}$ and $\Delta\vec{r}$ ($\cos \theta$).

Units of work:

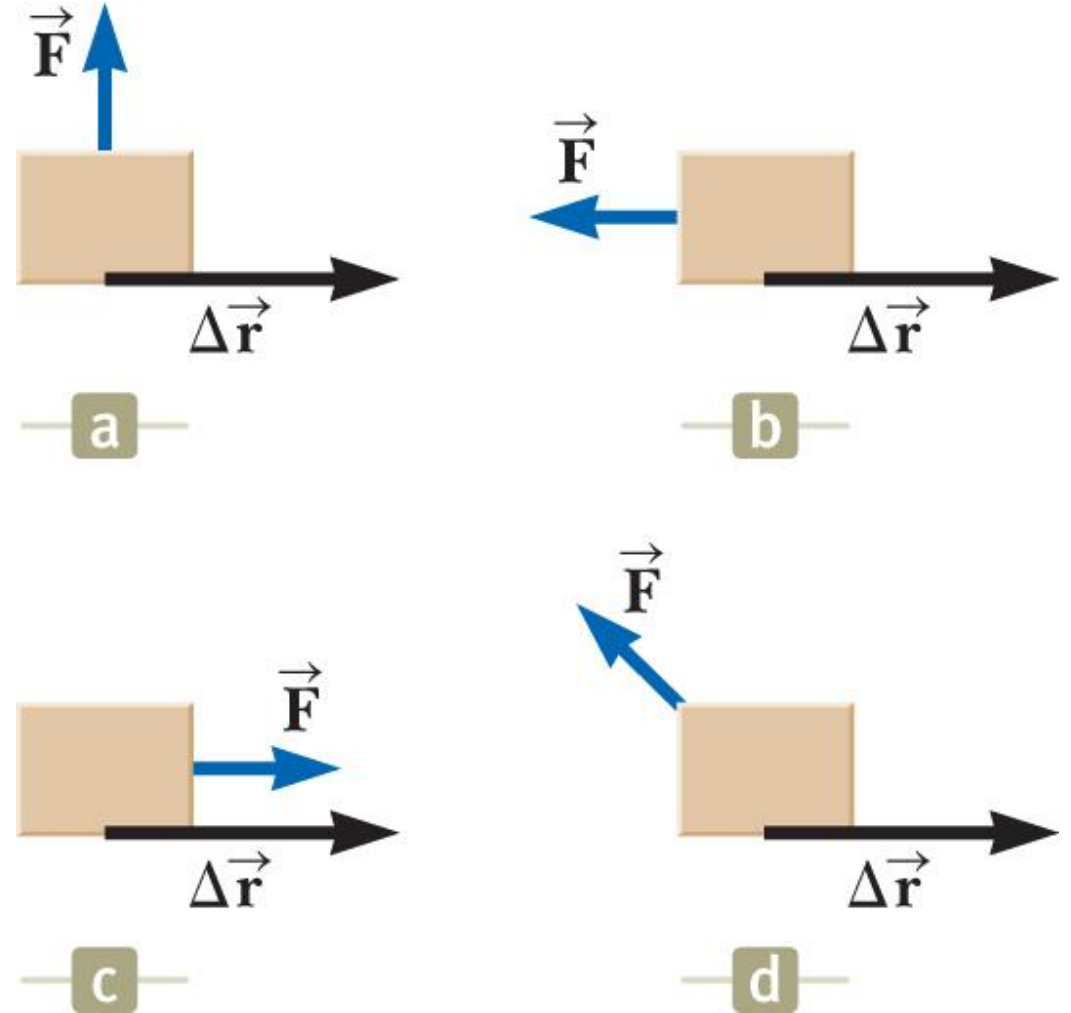
$$\text{N} \cdot \text{m} \equiv \text{J (joule)} = \text{kg} \cdot \text{m}^2/\text{s}^2$$

Rank the **work done** in the following!

The figure shows four situations in which a force is applied to an object.

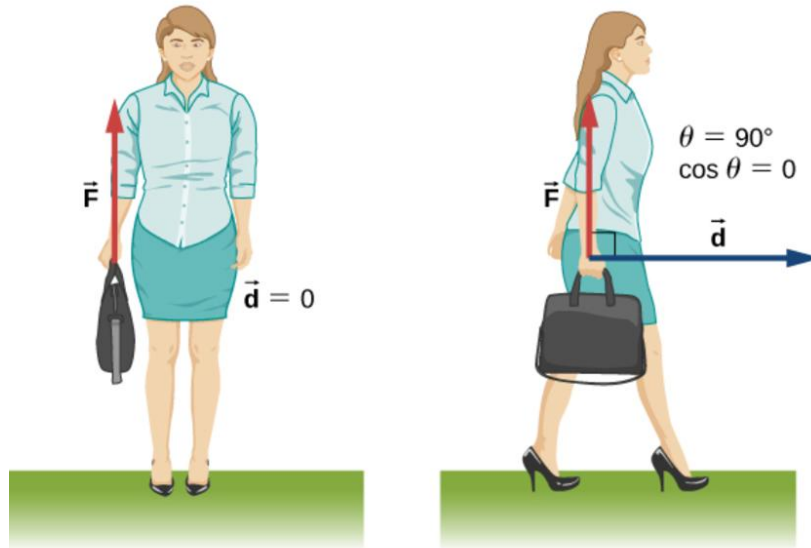
In all four cases, the **force** has the same magnitude, and the **displacement** of the object is to the right and of the same magnitude.

Rank the situations in order of the **work done** by the force on the object, from most positive to most negative.



Definition of Work

- Work meaning in physics is **distinct from** everyday meaning.
- Force does no work on an object if the object does not move through displacement: $\Delta r = 0 \rightarrow W = 0$.
- **Example 1** : A person holds a briefcase: **no work is done** because the displacement is zero.
- **Example 2** : The person walks horizontally while holding the briefcase: **no work is done** because $\cos\theta$ is zero.



$$W \equiv F \Delta r \cos \theta$$

Note: $\vec{d} \equiv \Delta r$

Work and Energy

- **Normal Force** : the normal force never does work under typical circumstances ($\theta = 90^\circ$):

$$dW_N = \vec{N} \cdot d\vec{r} = 0$$

- Note that if the displacement did have a relative component perpendicular to the surface, the object would leave the surface, and there would no longer be any normal contact force.
- **Frictional Force** : The work done by **kinetic friction** is negative, as kinetic friction is always opposite to the moving direction ($\theta = 180^\circ$)

$$W_{\text{fr}} = \int_A^B \vec{f}_k \cdot d\vec{r} = \int_A^B f_k dr \cos 180^\circ = -f_k \int_A^B |dr| = -f_k |l_{AB}|$$

Note: $\vec{d} \equiv \Delta r \equiv l_{AB}$

- The force of **static friction** does no work between two surfaces because there is never displacement between the surfaces.

Review: Scalar Product of Two Vectors

- **Scalar product (dot product)** of two vectors $\vec{A}$ and $\vec{B}$: $\vec{A} \cdot \vec{B}$

Scalar product: scalar quantity = product of magnitudes of two vectors and cosine of angle θ between them.

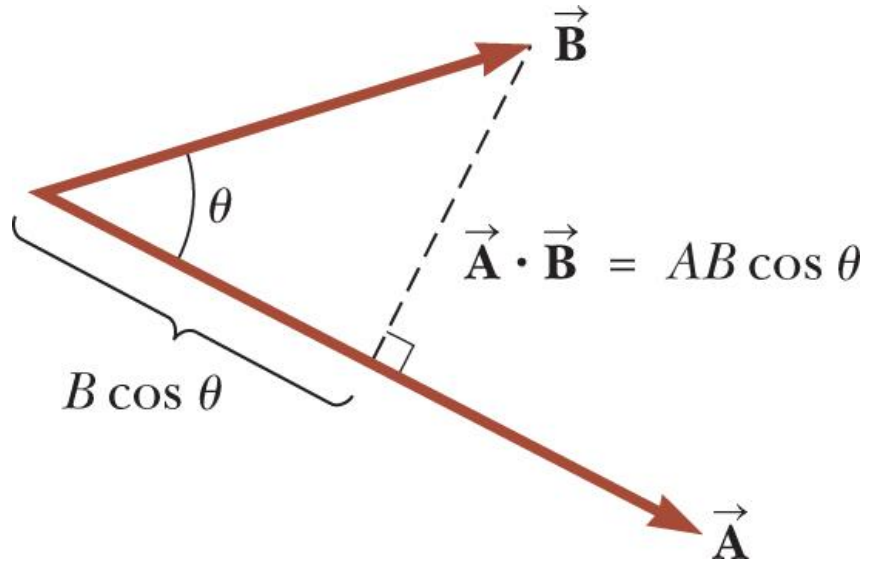
$$\vec{A} \cdot \vec{B} \equiv AB \cos \theta$$

Note: A and B need not have the same unit

$$W = F \Delta r \cos \theta = \vec{F} \cdot \Delta \vec{r}$$

Work : scalar product: $\vec{F} \cdot \Delta \vec{r} = F \Delta r \cos \theta$

Scalar Product of Two Vectors



$$\vec{A} \cdot \vec{B} \equiv AB \cos \theta$$

$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$$

$$\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$$

$$\vec{A} \perp \vec{B} (\theta = 90^\circ) \Rightarrow \vec{A} \cdot \vec{B} = 0$$

$$\vec{A} \parallel \vec{B} (\theta = 0^\circ) \Rightarrow \vec{A} \cdot \vec{B} = AB$$

$$\vec{A} \parallel \vec{B} (\theta = 180^\circ) \Rightarrow \vec{A} \cdot \vec{B} = -AB$$

Scalar Product of Two Vectors

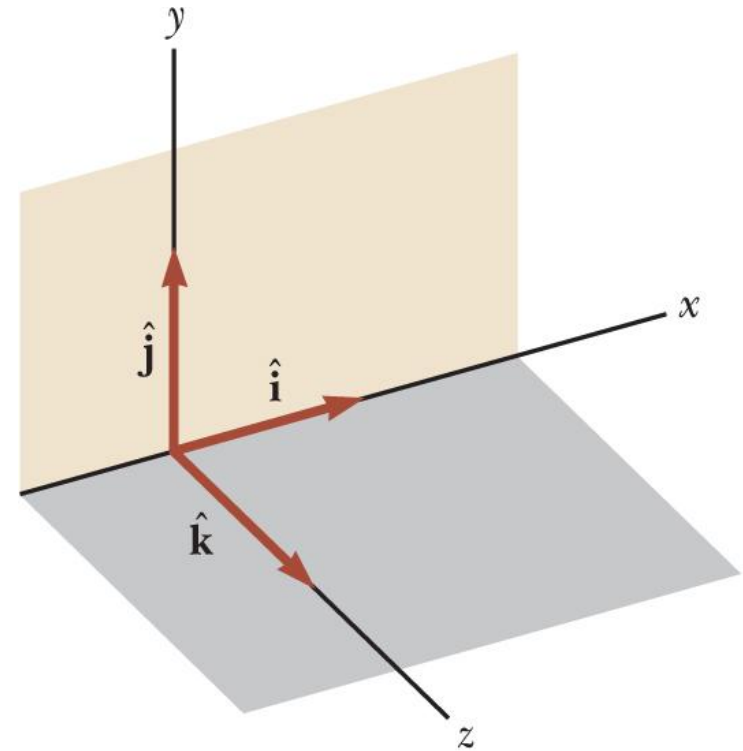
$$\vec{\mathbf{i}} \cdot \vec{\mathbf{i}} = \vec{\mathbf{j}} \cdot \vec{\mathbf{j}} = \vec{\mathbf{k}} \cdot \vec{\mathbf{k}} = 1 \quad \vec{\mathbf{i}} \cdot \vec{\mathbf{j}} = \vec{\mathbf{i}} \cdot \vec{\mathbf{k}} = \vec{\mathbf{j}} \cdot \vec{\mathbf{k}} = 0$$

$$\vec{\mathbf{A}} = A_x \vec{\mathbf{i}} + A_y \vec{\mathbf{j}} + A_z \vec{\mathbf{k}}$$

$$\vec{\mathbf{B}} = B_x \vec{\mathbf{i}} + B_y \vec{\mathbf{j}} + B_z \vec{\mathbf{k}}$$

$$\vec{\mathbf{A}} \cdot \vec{\mathbf{B}} = A_x B_x + A_y B_y + A_z B_z$$

$$\vec{\mathbf{A}} \cdot \vec{\mathbf{A}} = A_x^2 + A_y^2 + A_z^2 = A^2$$



Example#1: The scalar product

The vectors $\vec{\mathbf{A}}$ and $\vec{\mathbf{B}}$ are given by

$$\vec{\mathbf{A}} = 2\vec{\mathbf{i}} + 3\vec{\mathbf{j}} \text{ and } \vec{\mathbf{B}} = -\vec{\mathbf{i}} + 2\vec{\mathbf{j}}$$

(A) Determine the scalar product $\vec{\mathbf{A}} \cdot \vec{\mathbf{B}}$

$$\begin{aligned}\vec{\mathbf{A}} \cdot \vec{\mathbf{B}} &= (2\vec{\mathbf{i}} + 3\vec{\mathbf{j}}) \cdot (-\vec{\mathbf{i}} + 2\vec{\mathbf{j}}) \\ &= -2\vec{\mathbf{i}} \cdot \vec{\mathbf{i}} + 2\vec{\mathbf{i}} \cdot 2\vec{\mathbf{j}} - 3\vec{\mathbf{j}} \cdot \vec{\mathbf{i}} + 3\vec{\mathbf{j}} \cdot 2\vec{\mathbf{j}} \\ &= -2(1) + 4(0) - 3(0) + 6(1) = -2 + 6\end{aligned}$$

$$\boxed{\vec{\mathbf{A}} \cdot \vec{\mathbf{B}} = A_x B_x + A_y B_y = (2)(-1) + (3)(2) = 4}$$

Example#1: The scalar product

(B) Find the angle θ between $\vec{\mathbf{A}}$ and $\vec{\mathbf{B}}$.

$$A = \sqrt{A_x^2 + A_y^2} = \sqrt{(2)^2 + (3)^2} = \sqrt{13}$$

$$B = \sqrt{B_x^2 + B_y^2} = \sqrt{(-1)^2 + (2)^2} = \sqrt{5}$$

$$\cos \theta = \frac{\vec{\mathbf{A}} \cdot \vec{\mathbf{B}}}{AB} = \frac{4}{\sqrt{13}\sqrt{5}} = \frac{4}{\sqrt{65}}$$

$$\theta = \cos^{-1} \frac{4}{\sqrt{65}} = 60.3^\circ$$

Work is an Energy Transfer

- Energy $\rightarrow$ a physical quantity that is conserved.
- When physical process occurs $\rightarrow$ energy is **transferred** across the boundary of a system.
- If W is the work done on a system and W is **positive** $\rightarrow$ energy transferred **to the system**.
- If W is **negative** $\rightarrow$ energy is transferred **from the system**.
- If a system interacts with environment $\rightarrow$ interaction described as transfer of energy across system boundary, resulting in the change of energy stored in the system.

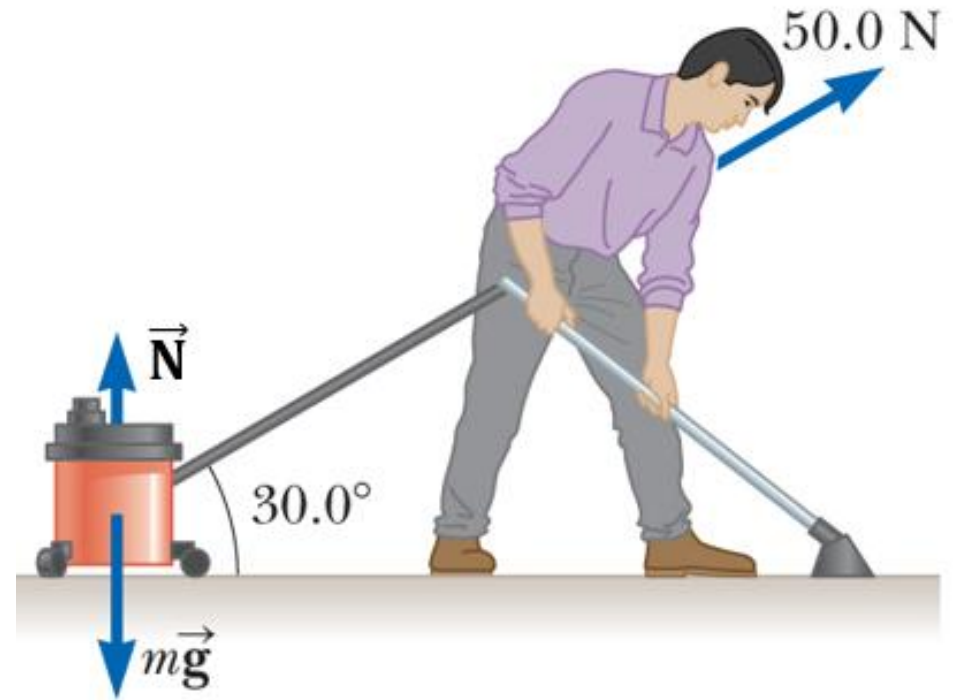
What's doing Work

- If there's a force with a component that is in the direction an object is moving, then work is being done
 - As long as they're not perpendicular
- Whatever is causing the **force** is doing the work
 - Friction
 - Gravity
 - Applied force (i.e. pushing something)
- We may have to draw **FBD** to determine the forces
 - If we want net work, then we need net force

Example#2 (Mr. Clean)

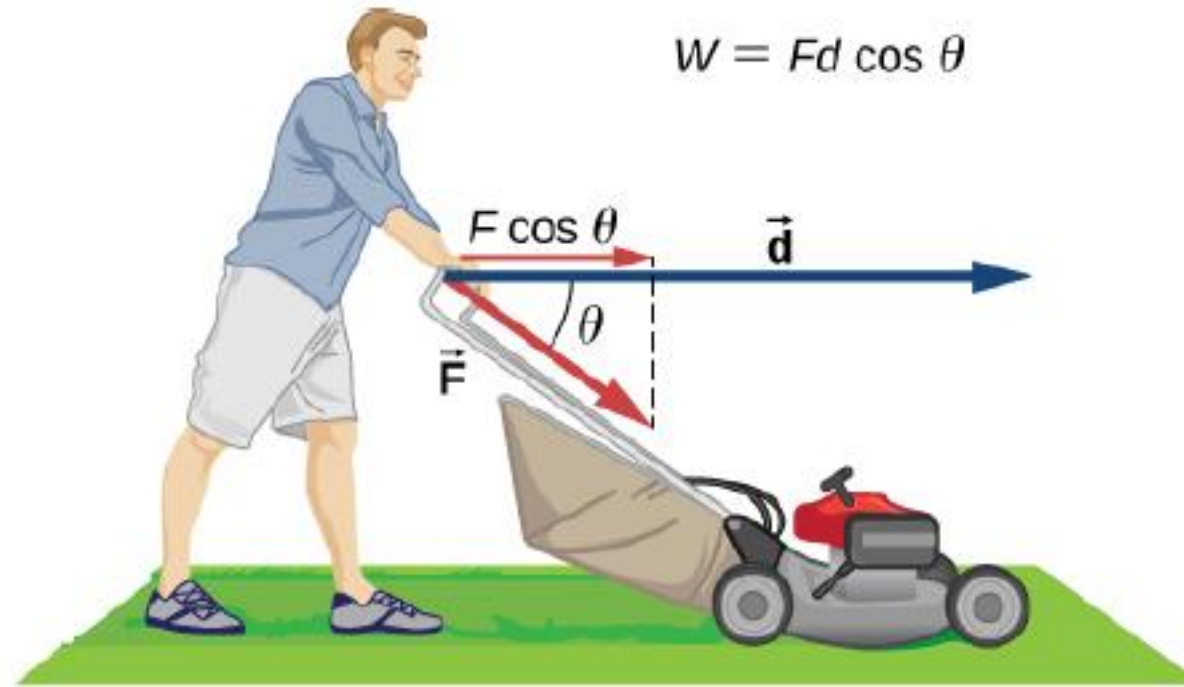
A man cleaning a floor pulls a vacuum cleaner with a force of magnitude $F = 50.0 \text{ N}$ at an angle of 30.0° with the horizontal. Calculate the work done by the **force** on the vacuum cleaner as the vacuum cleaner is displaced 3.00 m to the right.

$$\begin{aligned} W &= F \Delta r \cos \theta \\ &= (50.0 \text{ N})(3.00 \text{ m})(\cos 30.0^\circ) \\ &= \boxed{130 \text{ J}} \end{aligned}$$



Calculate the work done by the **Normal force**?

Example#2 Lawn Mower



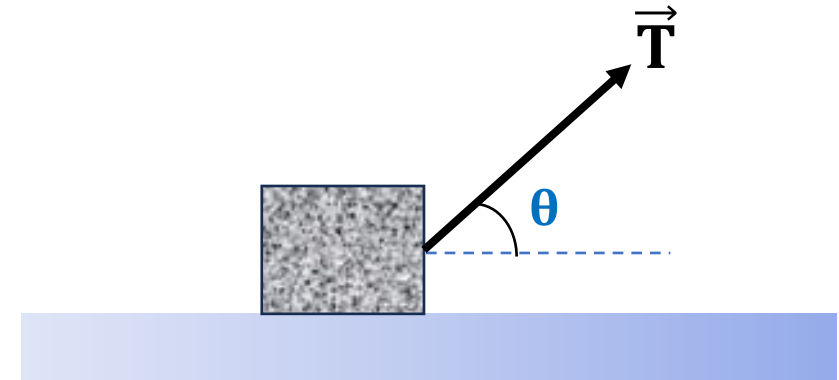
$$W \equiv F \Delta r \cos \theta$$

Note: $\vec{d} \equiv \Delta \vec{r}$

Example#3 (Concrete block)

A concrete block is pulled 7.0 m across a frictionless surface using a rope. The tension in the rope is 40 N; and the net work done on the block is 247 J. What **angle** does the rope make with the horizontal?

$$\theta = 28^\circ$$



Example#3 (Concrete block)

$$W = \vec{F} \cdot \vec{\Delta r}$$

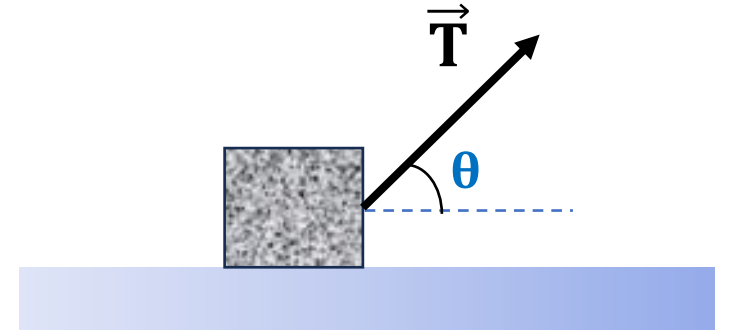
$$W_T = \vec{T} \cdot \vec{\Delta r} = T \Delta r \cos \theta$$

$$247 \text{ J} = (40 \text{ N})(7.0 \text{ m}) \cos \theta$$

$$\cos \theta = \frac{247}{(40)(7.0)} = 0.882$$

$$\theta = \cos^{-1}(0.882)$$

$$\theta = \dots - \text{degree}$$



Problem solving #25 (Textbook)

- (a) Calculate the work done on a 1500-kg elevator car by its cable to lift it 40.0 m at a constant speed, assuming friction averages 100 N.
- (b) What is the work done on the elevator car by the gravitational force in this process?
- (c) What is the total work done on the elevator car?

Answer: (a) 592 kJ

(b) – 588 kJ

(c) 0 J

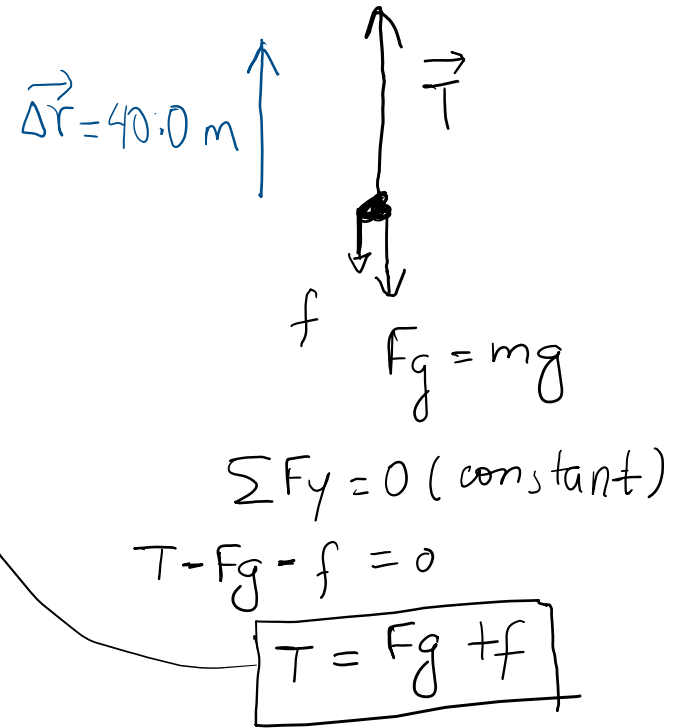
$$a) \quad W_T = \vec{T} \cdot \vec{\Delta r} = T \Delta r (\cos 0^\circ)$$

$$= [(F_g + f)] \Delta r (1)$$

$$= [(1500 \text{ kg})(9.80) + (100 \text{ N})] (40.0 \text{ m})$$

$$= 592000 \text{ J}$$

$$= 592 \text{ kJ}$$

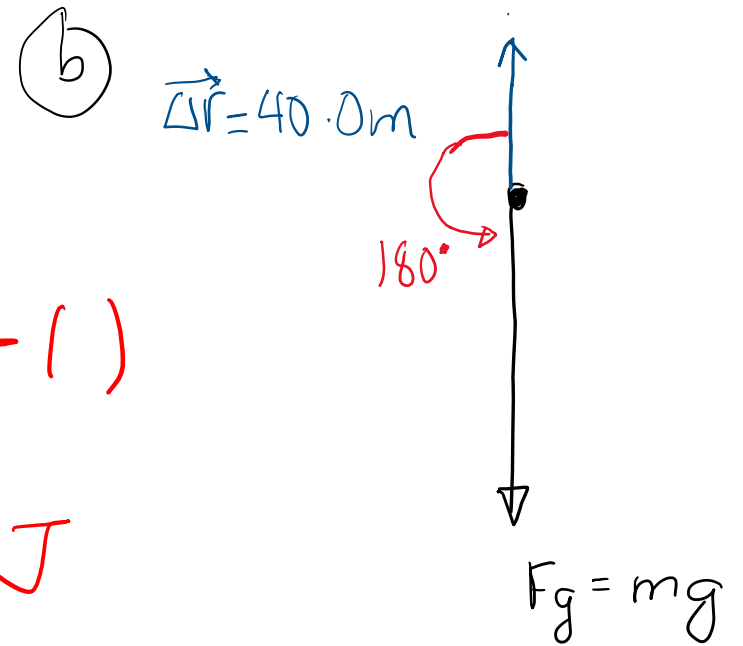


$$b) \quad W_g = |F_g| \cdot \vec{\Delta r}$$

$$= (mg) \text{ or } \cos 180^\circ$$

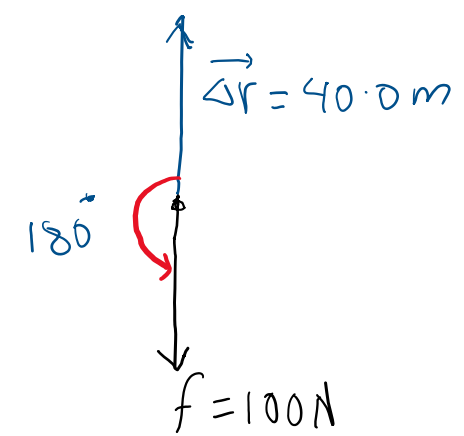
$$= (1500)(9.80)(40.0)(-1)$$

$$= -588000 \text{ J} = -588 \text{ kJ}$$



$$\begin{aligned}
 c) \quad W_{\text{Total}} &= W_T + W_g + W_f \\
 &= (592000\text{J}) + (-588000\text{J}) + (-4000\text{J}) \\
 &= (592000\text{J} - 592000\text{J})
 \end{aligned}$$

$$W_{\text{Total}} = 0\text{J}$$



$$\begin{aligned}
 W_f &= \vec{f} \cdot \vec{\Delta r} \\
 &= f \Delta r \cos 180^\circ \\
 &= (100\text{N})(40.0\text{m})(-1) \\
 &= -4000\text{J}
 \end{aligned}$$

ABCD card

An elevator supported by a single cable descends a shaft at a constant speed. The only forces acting on the elevator are the tension in the cable and the gravitational force. Which one of the following statements is true? **Ignore friction!**

- a. The magnitude of the work done by the tension force is larger than that done by the gravitational force.
- b. The magnitude of the work done by the gravitational force is larger than that done by the tension force.
- c. The work done by the tension force is zero joules.
- d. The work done by the gravitational force is zero joules.
- e. The net work done by the two forces is zero joules.

7.3 Work Done by a Varying Force

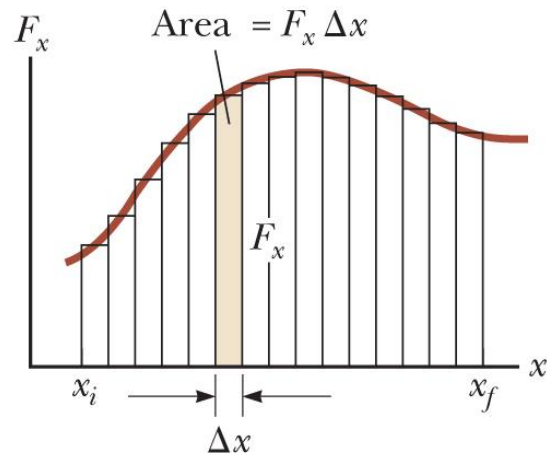
$$W \approx F_x \Delta x$$

$$W \approx \sum_{x_i}^{x_f} F_x \Delta x$$

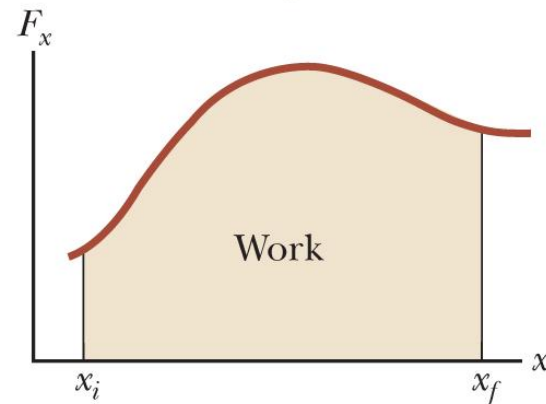
$$\lim_{\Delta x \rightarrow 0} \sum_{x_i}^{x_f} F_x \Delta x = \int_{x_i}^{x_f} F_x dx$$

$$W = \int_{x_i}^{x_f} F_x dx$$

The total work done for the displacement from x_i to x_f is approximately equal to the sum of the areas of all the rectangles.



The work done by the component F_x of the varying force as the particle moves from x_i to x_f is *exactly* equal to the area under the curve.



Problem solving #36 (Textbook)

How much work does the force $F(x) = \left(-\frac{2.0}{x}\right)$ N, do on a particle as it moves from $x = 2.0$ m to $x = 5.0$ m?

$$W = \int_{x_i}^{x_f} F_x dx$$

$$= \int_{2.0}^{5.0} \left(-\frac{2.0}{x}\right) dx$$

$$= (-2.0) \int_{2.0}^{5.0} \frac{1}{x} dx$$

$$W = (-2.0) \ln(x) \Big|_{2.0}^{5.0}$$

$$= (-2.0)[\ln(5.0) - \ln(2.0)]$$

$$= (-2.0)[1.60 - 0.693]$$

$$\boxed{W = -1.81 \text{ J}}$$

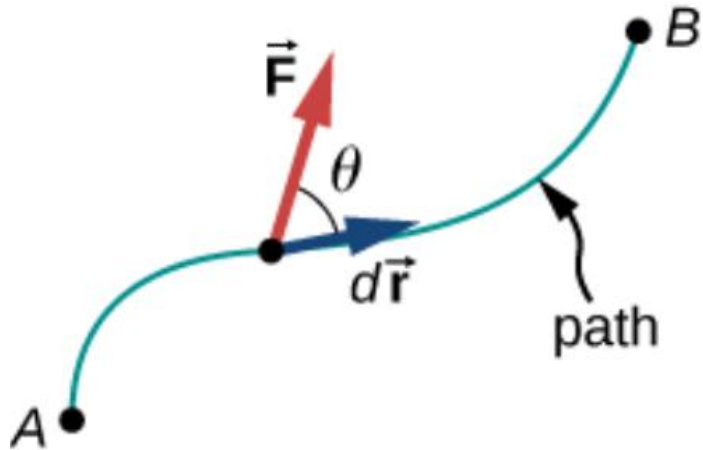
Work Done by a Varying Force

Work Done by a Force

The work done by a force is the integral of the force with respect to displacement along the path of the displacement:

$$W_{AB} = \int_{\text{path } AB} \vec{F} \cdot d\vec{r}.$$

7.2



$$dW = F_x dx + F_y dy + F_z dz.$$

The force acting on a particle and its infinitesimal displacement are shown at one point along the path between A and B . The infinitesimal work is the dot product of these two vectors; the total work is the integral of the dot product along the path.

Work Done by **Normal** and **Friction Force**

- **Normal Force** : the normal force never does work under typical circumstances ($\theta = 90^\circ$):

$$dW_N = \vec{N} \cdot d\vec{r} = 0$$

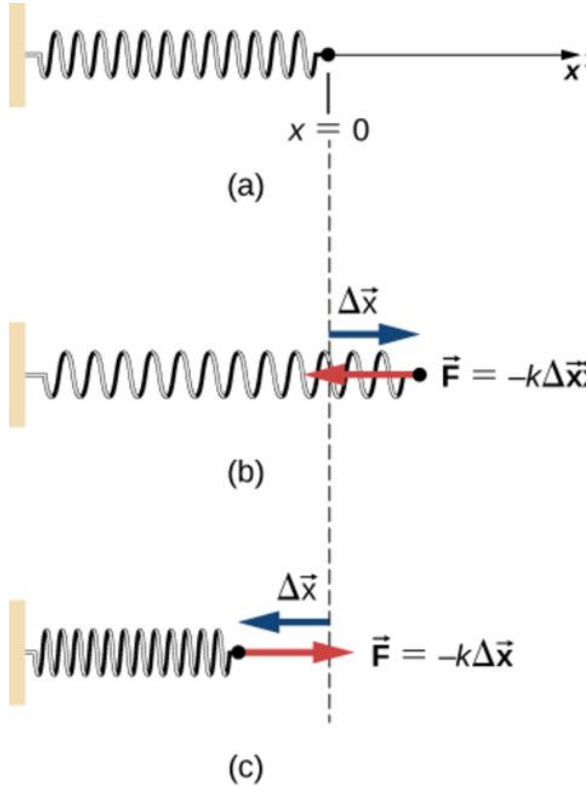
- Note that if the displacement did have a relative component perpendicular to the surface, the object would leave the surface, and there would no longer be any normal contact force.
- **Frictional Force** : The work done by **kinetic friction** is negative, as kinetic friction is always opposite to the moving direction ($\theta = 180^\circ$)

$$W_{\text{fr}} = \int_A^B \vec{f}_k \cdot d\vec{r} = \int_A^B f_k dr \cos 180^\circ = -f_k \int_A^B |dr| = -f_k |l_{AB}|$$

$$\text{Note: } \vec{d} \equiv \Delta r \equiv l_{AB}$$

- The force of **static friction** does no work between two surfaces because there is never displacement between the surfaces.

Work Done by a Spring Force



Hooke's law: $F_s = -kx$

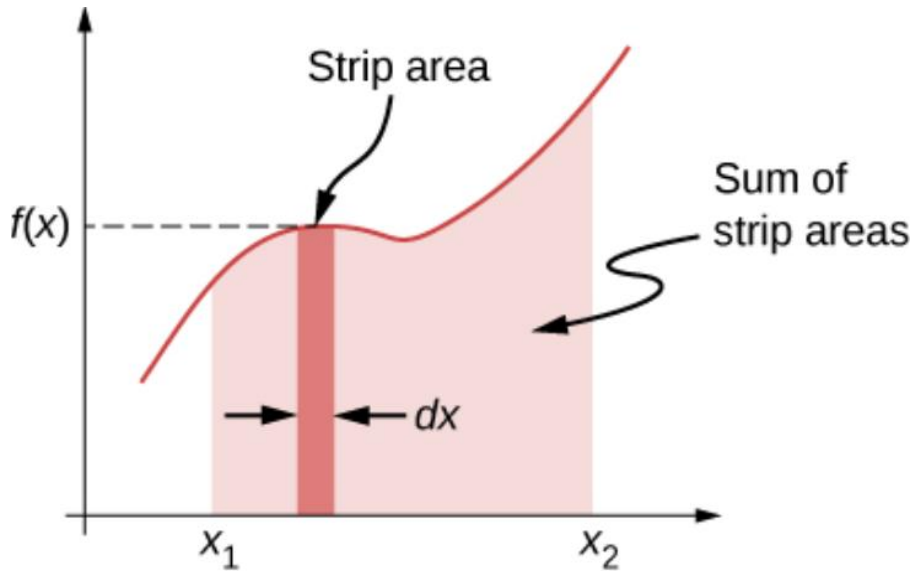
$$\vec{F}_s = F_s \vec{i} = -kx \vec{i}$$

k (spring constant): measure of the *stiffness* of a spring.

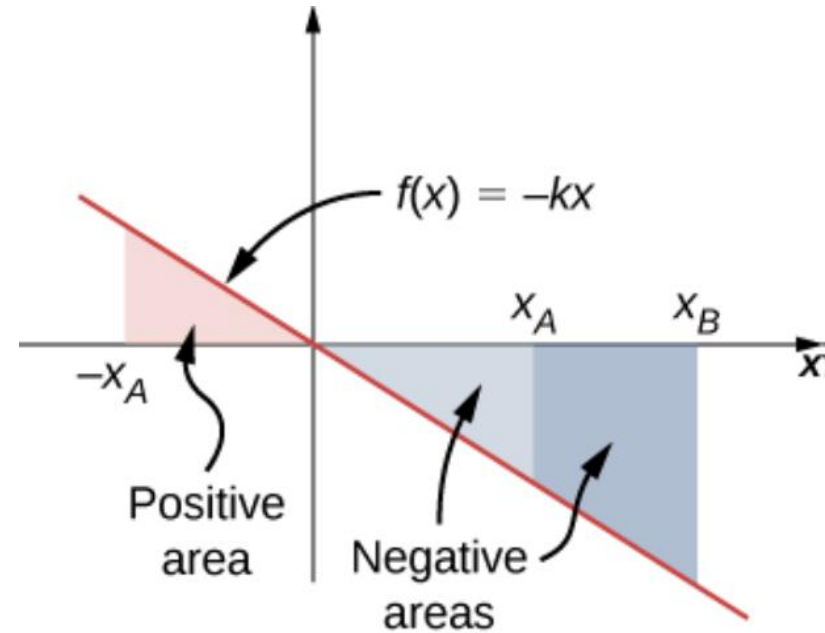
- The work done by spring force on any attached object is:

$$W_{\text{spring}, AB} = \int_A^B F_x dx = -k \int_A^B x dx = -k \frac{x^2}{2} \Big|_A^B = -\frac{1}{2} k (x_B^2 - x_A^2).$$

Calculating **Work** from a F-x Graph



A curve of $f(x)$ versus x showing the area of an infinitesimal strip, $f(x)dx$, and the sum of such areas, which is the integral of $f(x)$ from x_1 to x_2 .

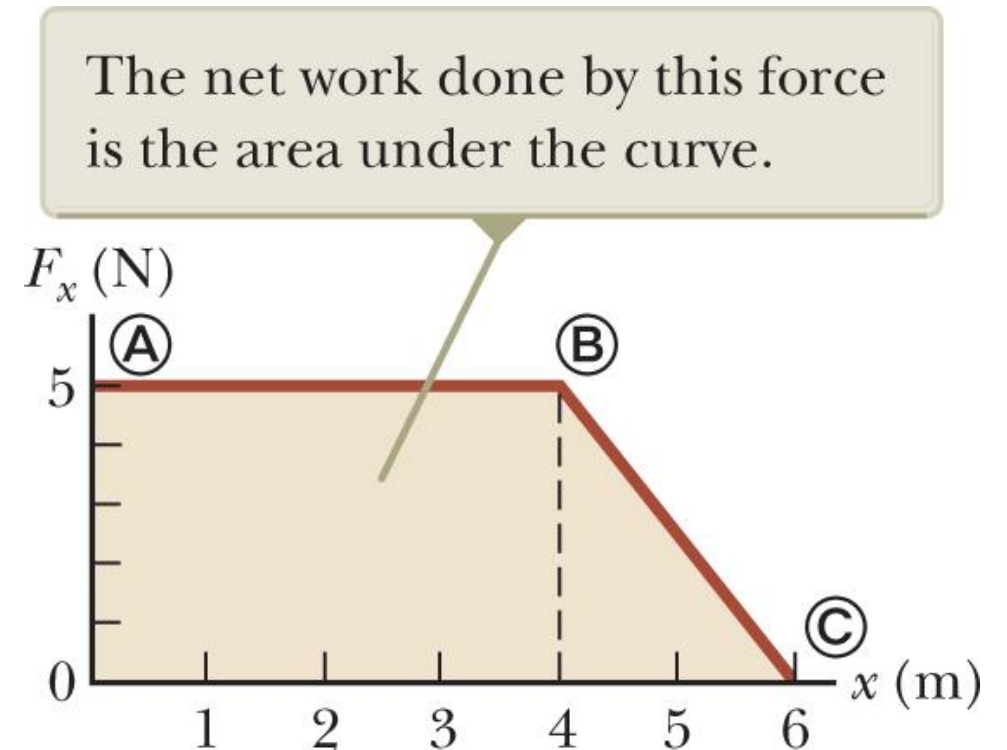


Curve of the spring force $f(x) = -kx$ versus x , showing areas under the line, between x_A and x_B , for both **positive** and **negative** values of x_A .

Calculating **Work** from a **F-x** Graph (Example)

A force acting on a particle varies with x as shown in the figure. Calculate the work done by the force on the particle as it moves from $x = 0$ to $x = 6.0$ m.

Answer = 25 J

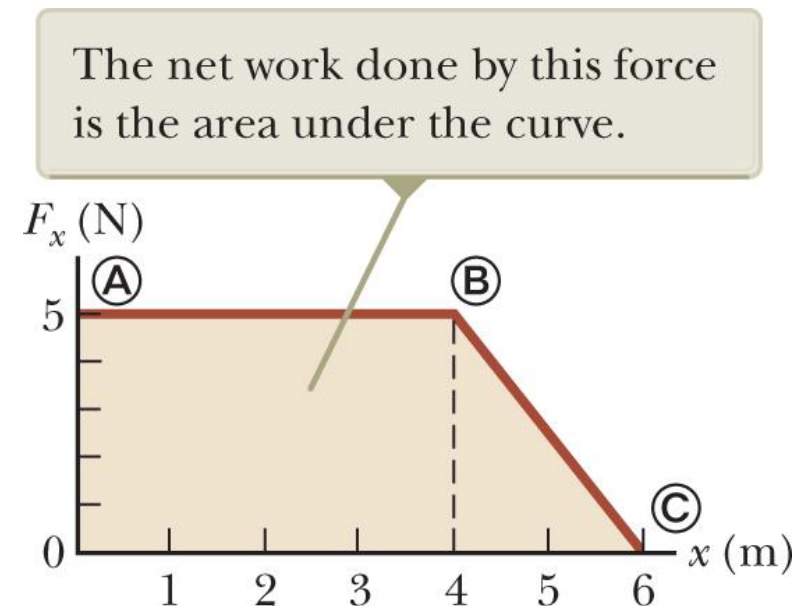


Calculating **Work** from a F-x Graph (Example)

$$W_{\text{A to B}} = (5.0 \text{ N})(4.0 \text{ m}) = 20 \text{ J}$$

$$W_{\text{B to C}} = \frac{1}{2}(5.0 \text{ N})(2.0 \text{ m}) = 5.0 \text{ J}$$

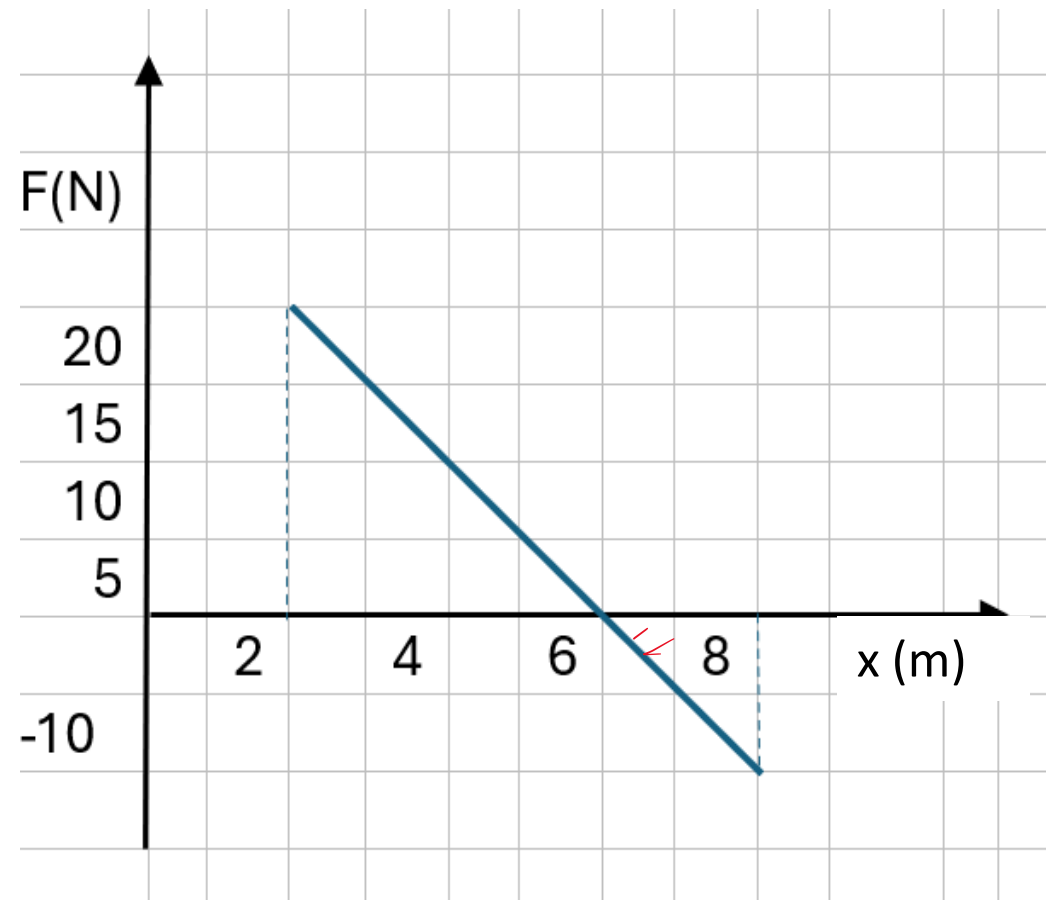
$$\begin{aligned} W_{\text{A to C}} &= W_{\text{A to B}} + W_{\text{B to C}} \\ &= 20 \text{ J} + 5.0 \text{ J} = \boxed{25 \text{ J}} \end{aligned}$$



Calculating **Work** from a F-x Graph (Example)

An object moving along the x axis is acted upon by a force F_x that varies with position as shown. How much work is done by this force as the object moves from $x = 2 \text{ m}$ to $x = 8 \text{ m}$?

Answer = 30 J



Calculating **Work** from a F-x Graph (Example)

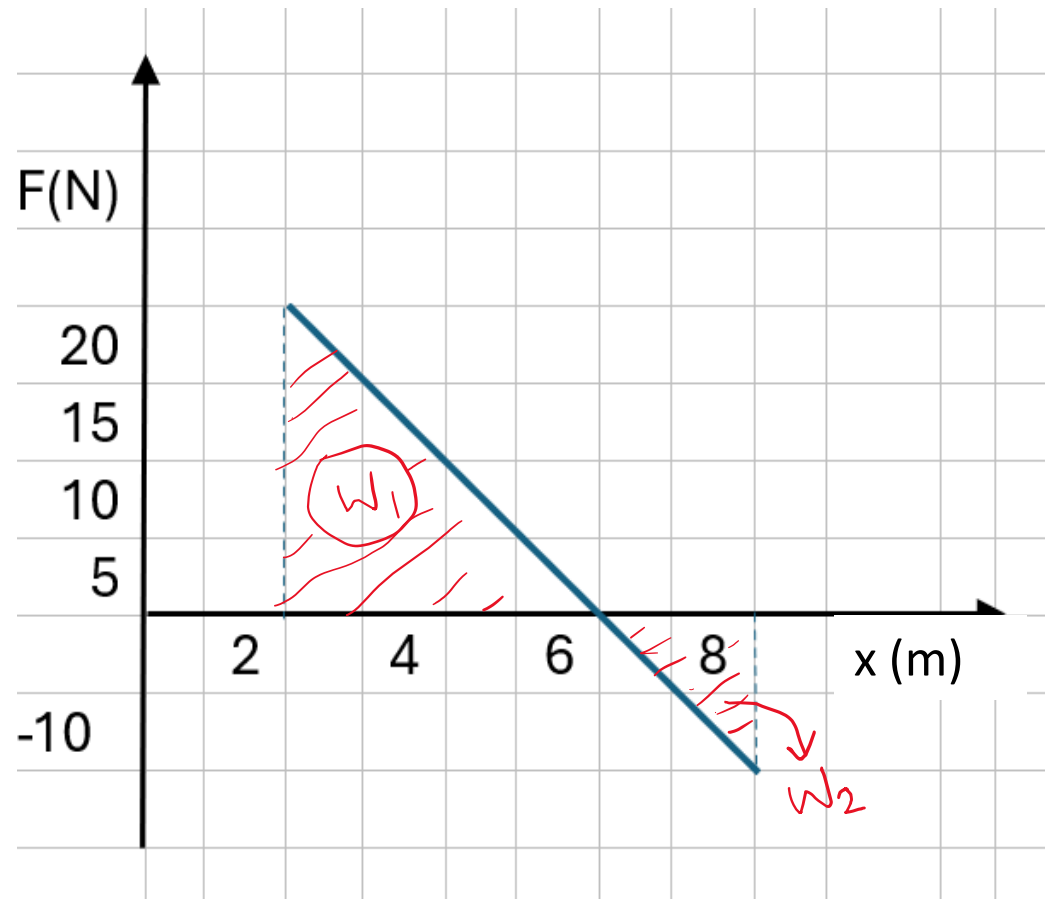
An object moving along the x axis is acted upon by a force F_x that varies with position as shown. How much work is done by this force as the object moves from $x = 2 \text{ m}$ to $x = 8 \text{ m}$?

Answer = 30 J

$$W_{\text{total}} = W_1 + W_2$$

$$W_1 = \frac{1}{2} (4 \text{ m}) (20 \text{ N}) = 40 \text{ Nm}$$

$$W_2 = \frac{1}{2} (2 \text{ m}) (-10 \text{ N}) = -10 \text{ Nm}$$



Conservative Force and Non-conservative force

- **Conservative forces** don't care about the path taken
 - Work is the same regardless
 - Ex. Gravity, elastic spring, electrostatic force
- **Non-conservative forces** do care about the path taken
 - Work is different depending on the path
 - Ex. Friction, air resistance, applied forces

ABCD card

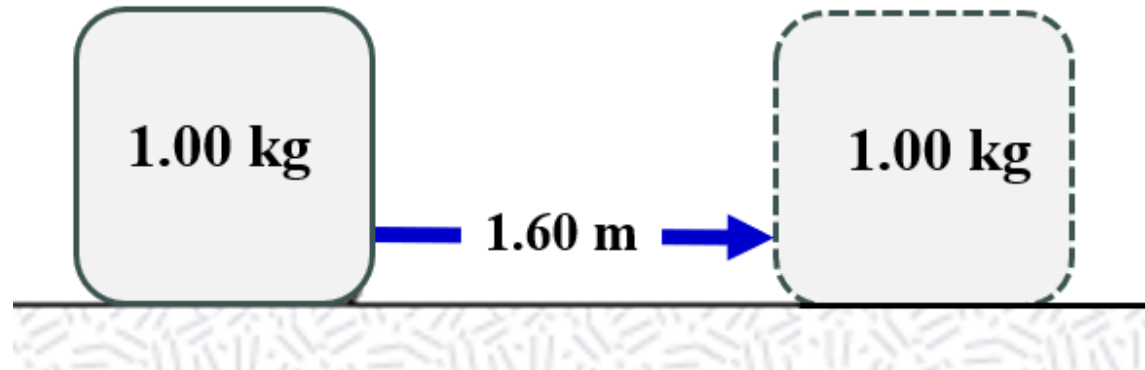
One car goes flying off a cliff and hits the ground at some point. A different car (same mass) takes the nice, safe road, which winds down the cliff and ends at the ground at the same point. In which case did gravity do the work greater?

- a. Car flying off cliff
- b. Car taking safe road
- c. Work done by gravity is the same
- d. All of the above

Work done by Friction (Example)

When a crate of 1.00 kg mass is dragged along a straight line for 1.60 m on a horizontal surface with a kinetic friction coefficient of 0.25, what is the **work done** by the **friction force** during this process?

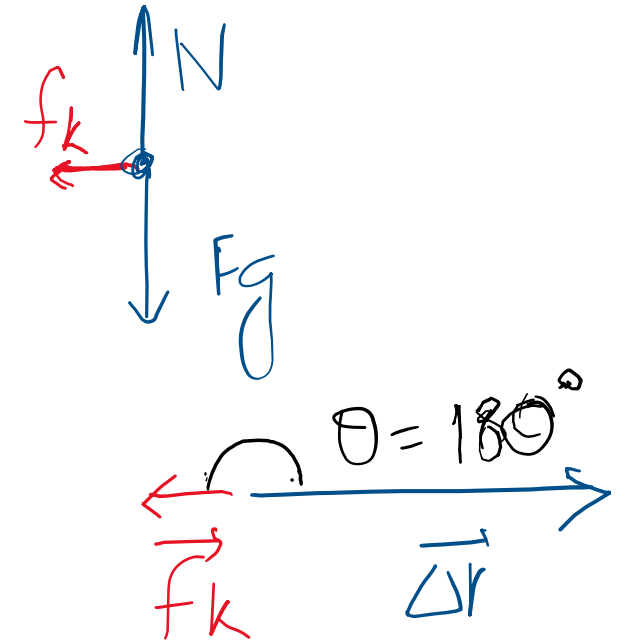
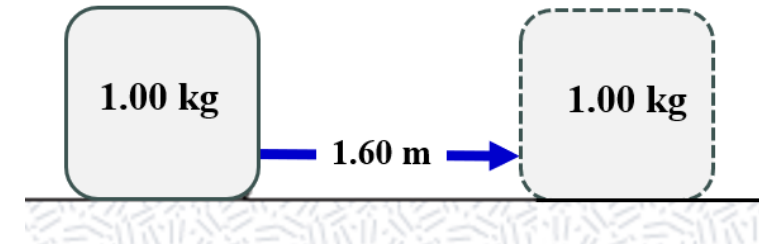
Answer: -3.92 J



Work done by Friction (Example)

$$\begin{aligned}W_{\text{friction}} &= \vec{f}_k \cdot \Delta \vec{r} \\&= f_k \Delta r \cos \theta \\&= f_k \Delta r \cos 180^\circ \\&= f_k \Delta r (-1) \\&= -f_k \Delta r \\&= -(\mu_k N) \Delta r \\&= -(\mu_k mg) \Delta r \\&= -(0.25)(1.00)(9.80)(1.60)\end{aligned}$$

$$W_{\text{friction}} = -3.92 \text{ J}$$

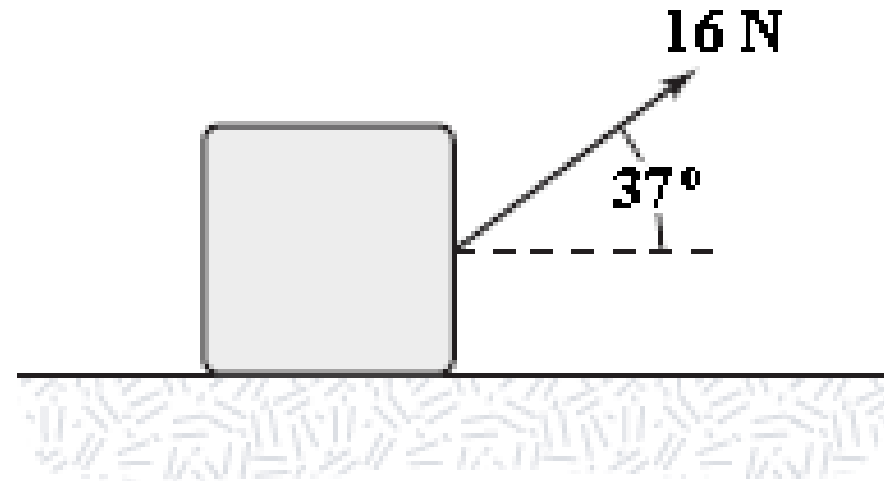


$$\vec{f}_k = \mu_k \vec{N}$$

Work done by Friction (Example)

A 3.0 kg block is dragged over a rough, horizontal surface by a constant force of 16 N acting at an angle of 37° above the horizontal level, as shown in the figure. The kinetic friction coefficient between the block and the surface is 0.25. The block had a displacement of 5.0 m. What is **work done** by the **friction force** during this displacement?

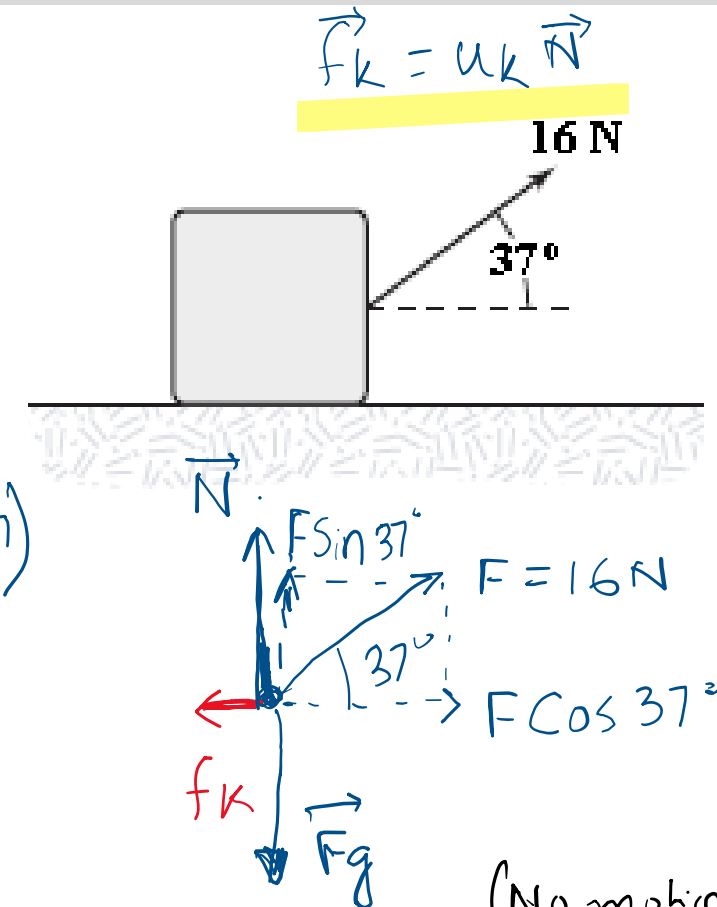
Answer: -25 J



Work done by Friction (Example)

$$\begin{aligned}W_{\text{friction}} &= f_k \Delta r \cos 180^\circ \\&= -f_k \Delta r \\&= -(u_k N) \Delta r \\&= -u_k (F_g - F \sin 37^\circ) \Delta r \\&= -(0.25) [(3.0)(9.80) - 16 \sin 37^\circ] (5.0 \text{ m})\end{aligned}$$

$$W_{\text{friction}} = -25 \text{ J}$$



(No motion along y-axis)

$$\sum F_y = 0$$
$$N + F \sin 37^\circ - F_g = 0$$
$$N = F_g - F \sin 37^\circ$$

7.4 Kinetic Energy

One possible result of doing work on system is, change in speed.

$$W_{net} = \int_{x_i}^{x_f} \sum F dx$$

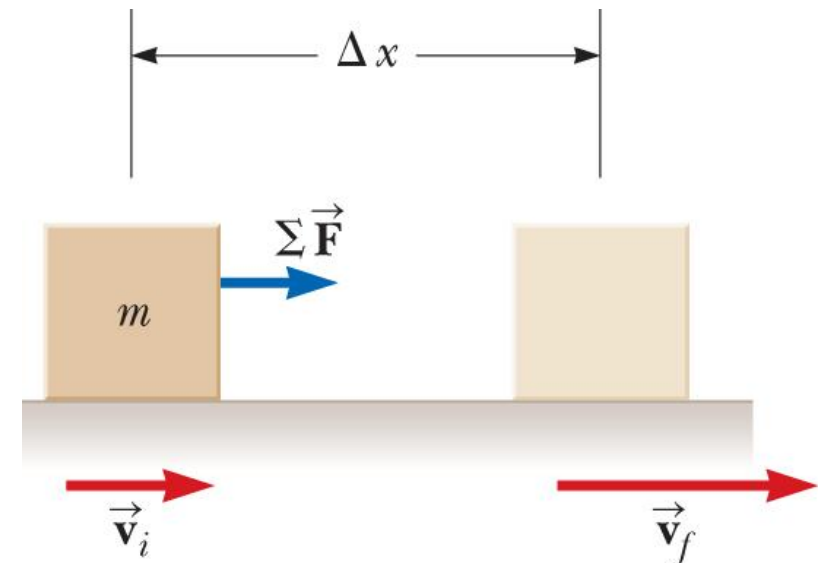
$$W_{net} = \int_{x_i}^{x_f} m a dx = \int_{x_i}^{x_f} m \frac{dv}{dt} dx = \int_{x_i}^{x_f} m \frac{dx}{dt} \frac{dv}{dx} dx = \int_{v_i}^{v_f} m v dv$$

Work Energy
Theorem

$$W_{net} = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2$$

Kinetic Energy: $K \equiv \frac{1}{2} m v^2$

$$\text{Unit} = \text{kg} \cdot \text{m}^2/\text{s}^2 = \text{J}$$



Kinetic Energy

Kinetic Energy

The kinetic energy of a particle is one-half the product of the particle's mass m and the square of its speed v :

$$K = \frac{1}{2}mv^2. \quad 7.6$$

- We then extend this definition to any system of particles by adding up the kinetic energies of all the constituent particles:

$$K = \sum \frac{1}{2}mv^2.$$

- Kinetic energy can also be expressed in terms of **momentum** ($\vec{p} = m\vec{v}$) and the kinetic energy of a particle can be expressed in terms of its mass instead of its mass and velocity:

$$K = \frac{1}{2}m\left(\frac{p}{m}\right)^2 = \frac{p^2}{2m}$$

kinetic energy and
Momentum.

7.5 Work-Energy Theorem

- Work done by the net force: $dW_{\text{net}} = \vec{\mathbf{F}}_{\text{net}} \cdot d\vec{\mathbf{r}}$.

$$dW_{\text{net}} = m \left(\frac{d\vec{\mathbf{v}}}{dt} \right) \cdot d\vec{\mathbf{r}} = m d\vec{\mathbf{v}} \cdot \left(\frac{d\vec{\mathbf{r}}}{dt} \right) = m \vec{\mathbf{v}} \cdot d\vec{\mathbf{v}},$$

$$\begin{aligned} W_{\text{net},AB} &= \int_A^B (mv_x dv_x + mv_y dv_y + mv_z dv_z) \\ &= \frac{1}{2} m \left| v_x^2 + v_y^2 + v_z^2 \right|_A^B = \left| \frac{1}{2} m v^2 \right|_A^B = K_B - K_A. \end{aligned}$$

Work-Energy Theorem

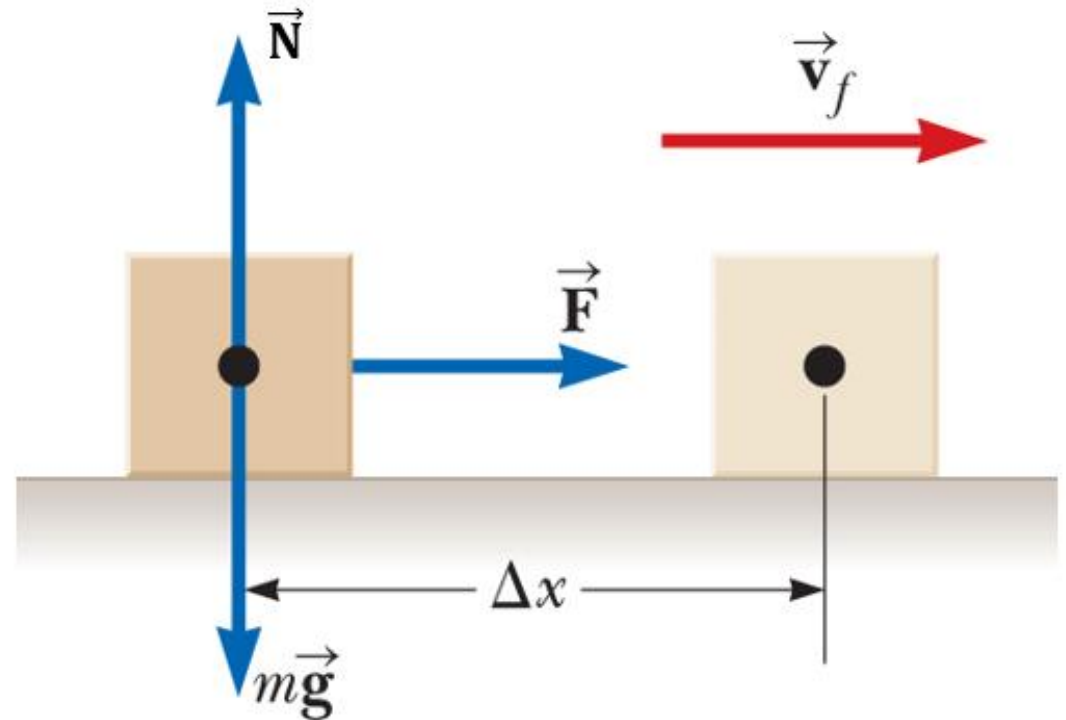
The net work done on a particle equals the change in the particle's kinetic energy:

$$W_{\text{net}} = K_B - K_A.$$

Work-Energy Theorem (Example)

A 6.0-kg block initially at rest is pulled to the right along a frictionless, horizontal surface by a constant horizontal force of magnitude 12 N.

Find the block's speed after it has moved through a horizontal distance of 3.0 m.



Work-Energy Theorem (Example)

Use the work–energy **Work Energy** theorem for the block:

$$W_{\text{net}} = \Delta K = K_f - K_i = \frac{1}{2}mv_f^2 - 0 = \frac{1}{2}mv_f^2$$

$$v_f = \sqrt{\frac{2W}{m}} = \sqrt{\frac{2F\Delta x}{m}}$$

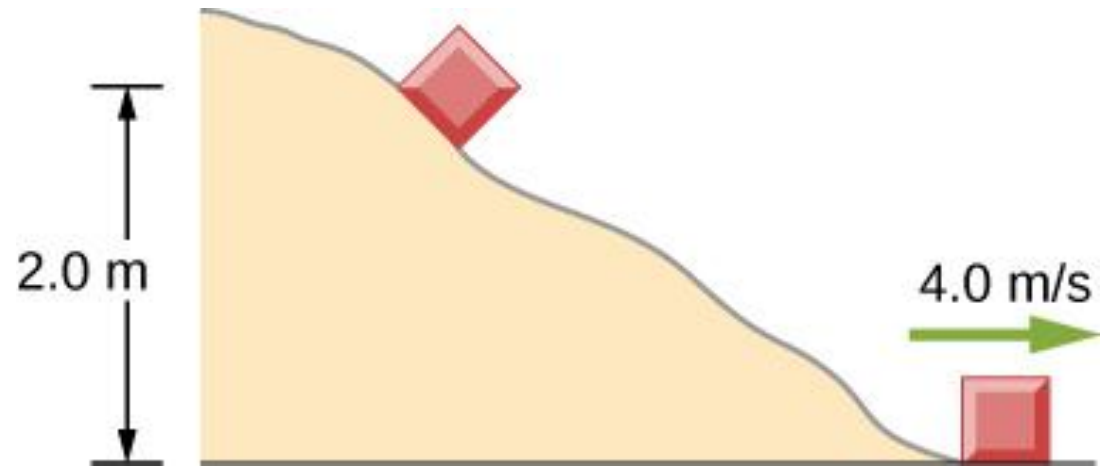
$$v_f = \sqrt{\frac{2(12 \text{ N})(3.0 \text{ m})}{6.0 \text{ kg}}} = \boxed{3.5 \text{ m/s}}$$

Note: You will get the same result using Newton's second law of motion.

Problem solving (Textbook #64)

When released, a 100-g block slides down the path shown below, reaching the bottom with a speed of 4.0 m/s. How much work does the force of friction do?

$$W_f = -1.2 \text{ J}$$



Problem solving (Textbook #64)

Work-Energy Theorem $W_{net} = \Delta K$

$$W_g + W_f = K_f - K_i$$

$$\text{friction} = \frac{1}{2} m(v_f)^2 - \frac{1}{2} m(v_i)^2$$

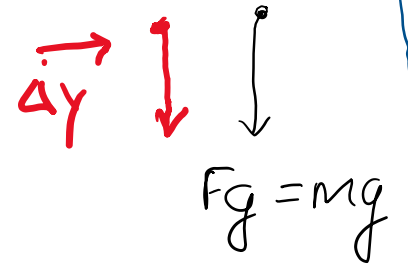
Force of gravity is conservative force, work done does not depend on the path.

$$\vec{F}_g \cdot \vec{\Delta y} + W_f = \frac{1}{2} (0.1 \text{ kg}) (4.0 \text{ m/s})^2$$

$$mg \Delta y \cos 0^\circ + W_f = 0.8 \text{ J}$$

$$\begin{aligned} W_f &= 0.8 \text{ J} - mg \Delta y \\ &= 0.8 \text{ J} - (0.1 \text{ kg}) (9.80 \text{ m/s}^2) (2.0 \text{ m}) \\ &= 0.8 \text{ J} - 1.96 \text{ J} \\ &= -1.16 \text{ J} \end{aligned}$$

$$W_f = -1.2 \text{ J}$$



$$W = F \cdot \Delta r$$

7.6 Power

- We express the relation between work done and the time interval involved in doing it, by introducing the concept of **power**.
- Since work can vary as a function of time, we first define **average power** as the work done during a time interval, divided by the interval:

$$P_{\text{avg}} = \frac{W}{\Delta t}$$

- Then, we define the **instantaneous power** (referred to as just **power**).

Power

Power is defined as the rate of doing work, or the limit of the average power for time intervals approaching zero,

$$P = \frac{dW}{dt}.$$

7.11

- Here we focus on work as energy transfer method → note that power is valid for *any* means of energy transfer.

Instantaneous Power

- Instantaneous power: limiting value of average power as $\Delta t \rightarrow$ zero:

$$P = \lim_{\Delta t \rightarrow 0} \frac{W}{\Delta t} = \frac{dW}{dt}$$

- dW = infinitesimal value of work done:

- For constant force $\rightarrow$ **instantaneous power** is: $dW = \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$

$$P = \frac{dW}{dt} = \frac{\vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}}{dt} = \vec{\mathbf{F}} \cdot \left(\frac{d\vec{\mathbf{r}}}{dt} \right) = \vec{\mathbf{F}} \cdot \vec{\mathbf{v}}$$

Power

SI unit of power: $1 \text{ W} = 1 \text{ J/s} = 1 \text{ kg} \cdot \text{m}^2 / \text{s}^3$

U.S. customary system (**horsepower**): $1 \text{ hp} = 746 \text{ W}$

Unit of energy (electrical transmission):

$$1 \text{ kWh} = (10^3 \text{ W})(3600 \text{ s}) = 3.60 \times 10^6 \text{ J}$$

$$(0.100 \text{ kW})(1.00 \text{ h}) = 0.100 \text{ kWh} = 3.60 \times 10^5 \text{ J}$$

Kilowatt hours (kW.h)

- One of the more common occurrences of Watts is with electrical devices
- On your power bill you are charged by the **kW.h**

To find the cost for a device

1. Take the Watts and convert it to kW (divide by 1000)
2. Multiply by hours used
3. Multiply this number by cost of kW·h

Problem solving (**Power**)

A child pulls a wagon with a horizontal force of 75 N. If the wagon moves horizontally a total of 42 m in 3.0 minutes. What is the power generated by the child?

Answer = 18 W

Problem solving (**Power**)

Given: Force = 75 N

distance = 32 m

time = 3.0 minutes = (3.0×60) seconds

$$\text{Power (P)} = \frac{W}{t}$$

$$= \frac{Fd \cos 0^\circ}{t}$$

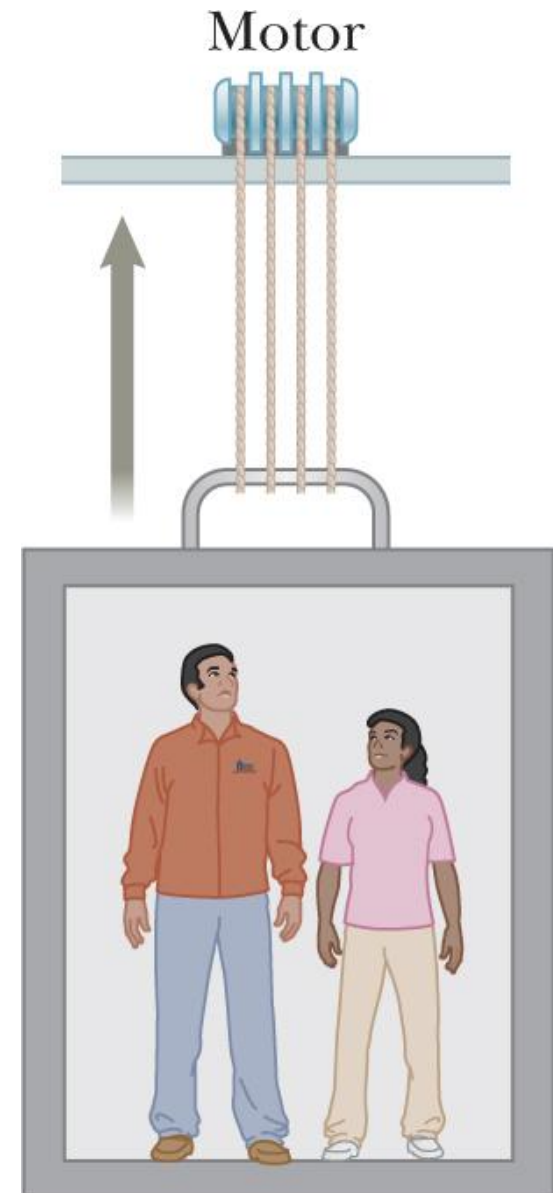
$$= \frac{(75 \text{ N}) (32 \text{ m})}{180 \text{ s}}$$

$$\boxed{P = 18 \text{ Watt}}$$

Power (Example)

An elevator car has a mass of 1600 kg and is carrying passengers having a combined mass of 200 kg. A constant **friction** force of 4000 N retards its motion.

(A) How much **power** must a motor deliver to lift the elevator car and its passengers at a constant speed of 3.00 m/s?



Power (Example)



$$\sum F_y = T - f - Mg = 0$$

$$T = Mg + f$$

$$P = \vec{T} \cdot \vec{v} = Tv = (Mg + f)v$$

$$\begin{aligned} P &= \left[(1800 \text{ kg})(9.80 \text{ m/s}^2) + (4000 \text{ N}) \right] (3.00 \text{ m/s}) \\ &= 6.49 \times 10^4 \text{ W} \end{aligned}$$

Power Delivered by Elevator Motor

(B) What power must the motor deliver at the instant the speed of the elevator is v if the motor is designed to provide the elevator car with an upward **acceleration** of 1.00 m/s^2 ?

Power Delivered by Elevator Motor

$$\sum F_y = T - f - Mg = Ma$$

$$T = M(a + g) + f$$

$$P = Tv = [M(a + g) + f]v$$

$$\begin{aligned} P &= [(1800 \text{ kg})(1.00 \text{ m/s}^2 + 9.80 \text{ m/s}^2) + 4000 \text{ N}]v \\ &= \boxed{(2.34 \times 10^4 \text{ N})v} \end{aligned}$$

$$P = (2.34 \times 10^4)(3.00 \text{ m/s}) = 7.02 \times 10^4 \text{ W}$$

Chapter 7 Summary

- Work done by a constant force: $W = F \Delta r \cos \theta = \vec{\mathbf{F}} \cdot \Delta \vec{\mathbf{r}}$
- Work done by a varying force: $W = \int_{x_i}^{x_f} F_x dx$
- Kinetic Energy: $K \equiv \frac{1}{2} m v^2$
- Work–Energy Theorem: $W_{\text{net}} = K_f - K_i = \Delta K$
- Power: $P = \frac{dW}{dt} = \frac{\vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}}{dt} = \vec{\mathbf{F}} \cdot \left(\frac{d\vec{\mathbf{r}}}{dt} \right) = \vec{\mathbf{F}} \cdot \vec{\mathbf{v}}$